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Fresnel diffraction due to phase gradient singularity

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In this Letter, a singularity in the phase gradient is introduced as a new origin of the diffraction from phase objects. The continuity in the amplitude and phase of the wave and the singularity in the phase gradient are assumed to describe this phenomenon. Fresnel diffraction of Fresnel double mirror and biprism are studied as practical examples, and the diffracted intensity distribution is calculated. We show that the intensity of the diffracted field varies with an almost constant period along the average propagation direction. Using this value and the fringe spacing of the interference pattern on a plane normal to the average propagation direction, the angle between two interfering beams and the wavelength of the incident light could be simultaneously derived. The diffraction pattern of the Fresnel double mirror and Fresnel biprism were obtained experimentally, which were in good agreement with the theoretical results. © 2018 Optical Society of America

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Standard diffraction is caused by abrupt changes in the amplitude of the electromagnetic wave due to the presence of an opaque object in the path of the wave. Based on recent studies, diffraction can also occur in response to abrupt changes in the phase of the wave [1–4]. Reflection from a physical step or the partial passing through a bounded transparent medium can cause Fresnel diffraction.

Although Fraunhofer diffraction has wide-reaching applications in optical metrology and spectroscopy, Fresnel diffraction has generally been known as a restrictive phenomenon, e.g., the restriction of the resolving power of an imaging system due to the diffraction limit. Moreover, the complexity of the calculation of Fresnel diffraction imposes limitations on its applicability. However, recently, researchers have shown that Fresnel diffraction from phase steps can be widely applied to optical metrology. Tavassoly *et al.* applied Fresnel diffraction from phase steps to measure the thickness of thin layers [5,6]. Alipour *et al.* applied the superposition of diffracted fields from

two similar hot and cold wires to obtain the temperature profile around the hot wire [7]. Khorshad *et al.* have used diffraction from phase steps for measuring nanometer displacements [8]. Fresnel diffraction has also been used for determining the refractive index of transparent materials with high precision [9–11]. A technique to specify the nonlinear refractive index of a sample is proposed in [12] and a wavemeter based on Fresnel diffraction from a 1D step with variable height is presented in [13]. In [14], the study of the diffraction pattern of a modified Fresnel double mirror (FDM), imposing an initial height between the mirrors, demonstrated that Fresnel diffraction is caused by the initial height.

The present work demonstrates a new origin of Fresnel diffraction, and we show that an intense change in the phase gradient of a wave causes Fresnel diffraction. In particular, the diffraction pattern was observed in the discontinuity of the phase gradient, despite the continuities in the amplitude and phase of the wave. In this study, the diffracted intensity distribution (DID) was obtained for an FDM and Fresnel biprism (FB) using Fresnel–Kirchhoff diffraction theory, and we demonstrate the singularity in the phase gradient causes a Fresnel diffraction pattern. The features of the diffraction pattern are studied in detail, and it is shown that these features depend on the FDM (FB) parameters and wavelength of the incident wave. The DID was experimentally investigated for FDM and FB and was consistent with the theoretical results. In addition, the experimental results were used to determine the wavelength and apex angle of FDM, simultaneously.

Figure 1 shows the schematic diagram of FDM with a coordinate system, such that the y and z axes coincide with the intersection of the mirrors and FDM's bisector, respectively. A linear source, perpendicular to the xz plane on S, is assumed at the distance R_0 from the origin, O. Using the Fresnel–Kirchhoff diffraction integral, the diffracted field is obtained at an arbitrary point, P, in the distance R_p from O:

$$U(p) = A\sqrt{\frac{-i}{\lambda}} \left[\int_{-\infty}^0 \frac{e^{-ik(r_0+r_p)}}{\sqrt{r_0 r_p}} dx + \int_0^{+\infty} \frac{e^{-ik(r'_0+r'_p)}}{\sqrt{r'_0 r'_p}} dx \right], \quad (1)$$

where A is the field amplitude and

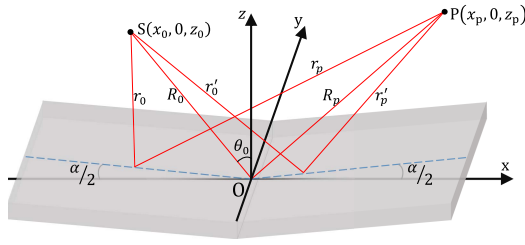


Fig. 1. Geometry to describe the Fresnel diffraction of light from the FDM. S is the point of a linear source on the xz plane, and P is the observation point. The intersection of the FDM is located on the y axis, and the mirrors make an angle of $\alpha/2$ with the xy plane.

$$\begin{aligned} r_i^2 &= (x - x_i)^2 + \left(-x \tan \frac{\alpha}{2} - z_i\right)^2, \\ r_i'^2 &= (x - x_i)^2 + \left(x \tan \frac{\alpha}{2} - z_i\right)^2, \end{aligned} \quad (2)$$

where $i = 0, p$. By considering the Fresnel approximation, the terms only up to the second order in the expansion of Eq. (2) are retained, so:

$$\begin{aligned} r_0 + r_p &= R_0 + R_p + \frac{1}{2R}x^2 - x\left(\beta - \gamma \tan \frac{\alpha}{2}\right), \\ r'_0 + r'_p &= R_0 + R_p + \frac{1}{2R}x^2 - x\left(\beta + \gamma \tan \frac{\alpha}{2}\right). \end{aligned} \quad (3)$$

In obtaining Eq. (3), the small terms $x^2 \tan^2 \frac{\alpha}{2}$ can be disregarded, and R , β , and γ are defined as follows:

$$\frac{1}{R} = \frac{1}{R_0} + \frac{1}{R_p}, \quad \beta = \frac{x_0}{R_0} + \frac{x_p}{R_p}, \quad \gamma = \frac{z_0}{R_0} + \frac{z_p}{R_p}. \quad (4)$$

By defining the quantities,

$$\begin{aligned} u &= R\left(\beta - \gamma \tan \frac{\alpha}{2}\right), \quad v = R_p + R_0 - \frac{R}{2}\left(\beta - \gamma \tan \frac{\alpha}{2}\right)^2, \\ u' &= R\left(\beta + \gamma \tan \frac{\alpha}{2}\right), \quad v' = R_p + R_0 - \frac{R}{2}\left(\beta + \gamma \tan \frac{\alpha}{2}\right)^2, \end{aligned} \quad (5)$$

Eq. (3) can be rewritten as follows:

$$\begin{aligned} r_0 + r_p &= v + \frac{1}{2R}(x - u)^2, \\ r'_0 + r'_p &= v' + \frac{1}{2R}(x - u')^2. \end{aligned} \quad (6)$$

Assuming that the width of FDM is w , the diffracted field is given by

$$\begin{aligned} U(p) &= A \sqrt{\frac{-i}{\lambda}} \left[e^{-ikv} \int_{-\frac{w}{2}}^0 \frac{e^{\frac{ik}{2R}(x-u)^2}}{\sqrt{r_0 r_p}} dx + e^{-ikv'} \int_0^{\frac{w}{2}} \frac{e^{\frac{ik}{2R}(x-u')^2}}{\sqrt{r'_0 r'_p}} dx \right] \\ &= A \sqrt{\frac{-i}{2(R_0 + R_p)}} e^{-ikv} [l(\chi_2, \chi_1) + e^{-ik(v-v')} l(\chi'_2, \chi'_1)]. \end{aligned} \quad (7)$$

In Eq. (7), l is defined as

$$l(\chi_j, \chi_i) = [C(\chi_j) - C(\chi_i)] - i[S(\chi_j) - S(\chi_i)], \quad (8)$$

where C and S are the Fresnel integrals [15]. χ_i and χ'_i ($i = 1, 2$) are obtained from the following equations:

$$\begin{aligned} \chi_1 &= -\sqrt{\frac{2}{\lambda R}} \left(u + \frac{w}{2}\right), \quad \chi_2 = -\sqrt{\frac{2}{\lambda R}} u, \\ \chi'_1 &= -\sqrt{\frac{2}{\lambda R}} u', \quad \chi'_2 = -\sqrt{\frac{2}{\lambda R}} \left(u' - \frac{w}{2}\right). \end{aligned} \quad (9)$$

The DID at point P is obtained by

$$I = I_0 |l(\chi_2, \chi_1) + e^{-ik(v-v')} l(\chi'_2, \chi'_1)|^2, \quad (10)$$

where $I_0 = \frac{|A|^2}{2(R_0 + R_p)}$. Notably, Eq. (10) can be applied when the axial distance between two mirrors is $m \frac{\lambda}{2}$, where m is an integer.

Using Eq. (10), the DID for an FDM with the angle $\alpha = 0.35^\circ$ was calculated. The FDM was illuminated by a monochromatic plane wave ($R_0 \Rightarrow \infty$ and $\lambda = 632.8$ nm) propagating along the z axis. Due to symmetry of the linear detector, with a size of 2 mm and pixel size of 2.2 μm , was assumed parallel to the x axis and scanning was along the z axis. Figure 2(a) shows the results for $z_p = 1$ mm to $z_p = 300$ mm with steps of $\Delta z = 0.1$ mm. It is noted that in addition to the FDM interference fringes, the diffraction pattern due to the singularity in the phase gradient was clearly observed. The specified region in Fig. 2(a) is magnified in Fig. 2(b), which demonstrates the intensity variation along the z axis caused by the Fresnel diffraction from the FDM. By considering only the interference of the two reflected waves, a constant intensity profile along the z axis was expected. Figure 2(c) shows the intensity profile along the z axis, where the intensity varied with an almost constant period, although the visibility was reduced by increasing the distance from the FDM. Figures 2(d)–2(g) show the intensity profiles along the x axis for $z_p = 13, 30, 47$, and 81 mm, respectively. Advancing along the z axis, the geometrical overlap of the two reflected beams was extended, which caused an increase in the interference area

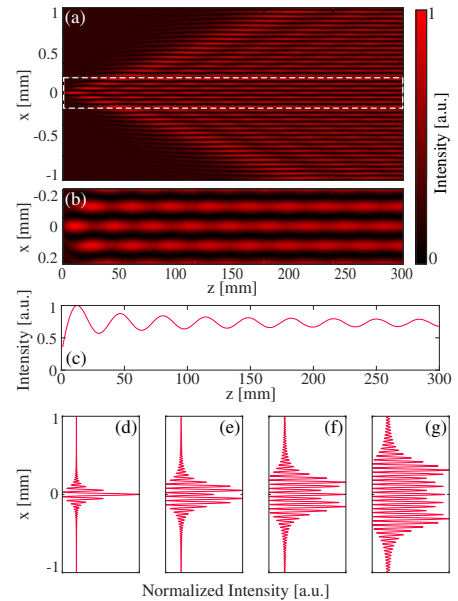


Fig. 2. Calculated diffracted intensity from the FDM: (a) intensity distribution on the xz plane; (b) the magnification of the specified region in (a); (c) intensity profile along the z axis; and intensity profiles along the x axis at: (d) $z_p = 13$ mm, (e) $z_p = 30$ mm, (f) $z_p = 47$ mm, and (g) $z_p = 81$ mm.

and therefore the number of interference fringes. The distinct differences at the maximum of the interferograms were interpreted by considering the diffraction effects. Outside of the geometrical overlapping area, there was no interference pattern, and only the diffraction effects were observed. Moreover, the visibility of the fringes approached zero for positions away from the overlapping area. The behavior of the intensity along the z axis was studied by imposing $x, y = 0$ in Eq. (10) as follows:

$$I = 4I_0 \left[\left(C(\chi_2) + \frac{1}{2} \right)^2 + \left(S(\chi_2) + \frac{1}{2} \right)^2 \right]. \quad (11)$$

The distance between the maximum intensities along the z axis is given by

$$\Lambda_z = \frac{\lambda}{2 \tan^2 \frac{\alpha}{2}}. \quad (12)$$

On the other hand, the separation of the interference fringes is given by the well-known expression:

$$\Lambda_x = \frac{\lambda}{2 \sin \alpha}, \quad (13)$$

where the angle between two interfering beams is 2α . Solving Eqs. (12) and (13), and assuming $\alpha \ll 1$ provides λ and α , simultaneously:

$$\alpha = 4 \frac{\Lambda_x}{\Lambda_z}, \quad \lambda = 8 \frac{\Lambda_x^2}{\Lambda_z}. \quad (14)$$

Equation (14) simultaneously provides the wavelength of the incident wave and the angle between the two mirrors. More specifically, unlike traditional FDM interferometry, the angle between mirrors could be derived independently of the wavelength.

The diffraction due to the phase gradient singularity is also observable in an FB. Because no phase singularity was applied to the transmitted light from the FB, the diffraction pattern was only due to the phase gradient singularity. The equations of the FDM will now be derived to obtain the DID from the FB. Considering that the FB is illuminated along the z axis (Fig. 3), r_0 and r'_0 are defined as follows:

$$\begin{aligned} r_0 &= R_0 + n \left(\frac{w}{2} + x \right) \tan \frac{\alpha}{2} & x < 0, \\ r'_0 &= R_0 + n \left(\frac{w}{2} - x \right) \tan \frac{\alpha}{2} & x \geq 0, \end{aligned} \quad (15)$$

where α is the FB's apex angle deviation from 180° . With simple calculations, it follows that the DID can be derived from Eq. (10), assuming that in Eq. (5) γ is replaced with $\Gamma = \gamma - n$, and

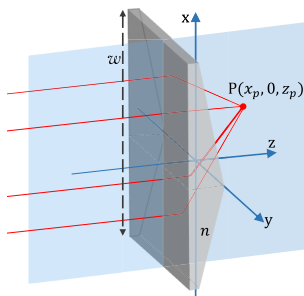


Fig. 3. Geometry used to describe diffraction from the FB.

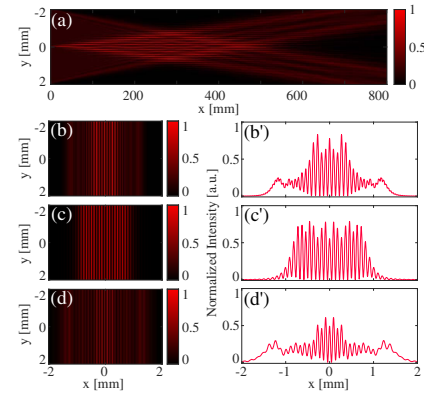


Fig. 4. Calculated diffracted intensity for the FB with $\alpha = 0.8^\circ$: (a) DID on the xz plane; (b), (c), and (d) DIDs on $z = 160, 320$, and 500 mm planes, respectively; (b'), (c'), and (d') are the corresponding profiles of (b), (c), and (d), respectively.

$$v - v' = 2x_p(n - 1) \tan \frac{\alpha}{2}. \quad (16)$$

The proposed formulation applies to any optical phase object with constant and singular phase gradient. Optical phase objects with variable and singular phase gradients can be considered as a combination of two optical objects, variable phase gradient object, and singular phase gradient object, and this formulation is used to describe the DID of them.

The DID is calculated for an FB with $\alpha = 0.8^\circ$ and $w = 4$ mm. Figure 4(a) shows the calculated DID in the xz plane. Figures 4(b)–4(d) show the DIDs for $z = 160, 320$, and 500 mm planes, respectively, and the corresponding profiles are presented in Figs. 4(b')–4(d'), respectively.

A schematic diagram of the experimental setup for the FDM and FB is shown in Figs. 5(a) and 5(b), respectively. A He–Ne laser beam was expanded and collimated by a spatial filter (SF) and lens (C). The DID was recorded by a camera (BaslerdaA2500-14um, 2592×1944 resolution, and $2.2 \mu\text{m}$ pixel pitch). For the case of the FDM, an xy two-axes translation stage (TS) was used to decrease the in-plane gap, and the axial distance between the two mirrors was made as small as possible. In addition, two kinematic mounts were used to align the apex angle of the FDM, and a beam splitter (BS) was used to direct the diffracted beam toward the camera. The camera was mounted on a linear stage providing axial displacements (Z-Scan) for $z = 5$ to 250 mm with the resolution of 0.1 mm.

The recorded DID of FDM on the xz plane is shown in Fig. 6(a). In Fig. 6(a'), the red dots show the corresponding

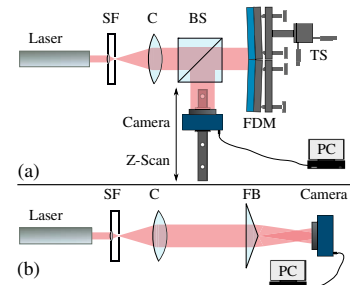


Fig. 5. Schemes of the experimental setups to investigate the DID from (a) FDM and (b) FB.

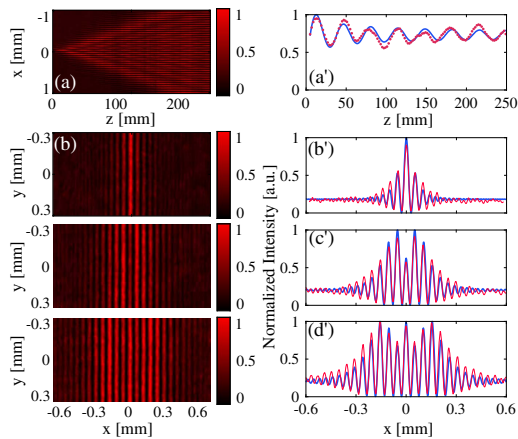


Fig. 6. DID of FDM and its profiles compared with theoretical results (red curves: experimental results; blue curves: theoretical results): (a) the DID of the FDM in xz plane; (a') the intensity profile along the z axis; (b), (c), and (d) the DIDs; (b'), (c'), and (d') their corresponding profiles for $z = 13, 30$, and 47 mm, respectively.

profile along the z axis, and the blue solid line shows the theoretical result for the same parameters. The experimental DIDs for $z = 13, 30$, and 47 mm are shown in Figs. 6(b)–6(d), respectively, and their corresponding profiles are shown in Figs. 6(b')–6(d') by red curves. The blue curves indicate the theoretical results with the similar conditions.

Using the experimental results, the separation of interference fringes and the separation between maximum intensities along the z axis are derived as $\Lambda_x = 51.74 \pm 0.03 \mu\text{m}$ and $\Lambda_z = 33.82 \pm 0.05 \text{ mm}$. The total uncertainty consists of systematic and statistical uncertainties. The systematic uncertainties for determining the Λ_x and Λ_z are calculated using $d\Lambda_x = \frac{dx}{M}$ and $d\Lambda_z = \frac{dz}{N}$, where $M = 100$ and $N = 7$ are the number of intensity maxima, and $dx = 2.2 \mu\text{m}$ and $dz = 100 \mu\text{m}$ are the measurement resolution along the x and z axes, respectively. The statistical uncertainties of measurement for Λ_x and Λ_z are $0.02 \mu\text{m}$ and 0.04 mm , respectively, which are calculated using several interference fringes and intensity spacing along the z axis. Consequently, the wavelength and the apex angle of FDM are obtained as $\lambda = 633 \pm 1.7 \text{ nm}$ and $\alpha = 0.3506 \pm .0007^\circ$.

The systematic uncertainties for α and λ are $\frac{d\alpha}{\alpha} + \frac{d\lambda}{\lambda}$ and $\frac{2d\alpha}{\alpha} + \frac{d\lambda}{\lambda}$, respectively, where L is the distance between two points with the maximum intensity along the z axis. Comparing with conventional interferometry, $\frac{d\alpha}{\alpha}$ is an additional term, which for the particular parameters $dx = 2 \mu\text{m}$, $w = 10 \text{ mm}$, $dz = 10 \mu\text{m}$, and $L = 500 \text{ mm}$, is one order of magnitude smaller than the first term and could therefore be neglected.

The DID and corresponding profiles for the FB with $\alpha = 2^\circ$ at distances $z = 70, 100$, and 160 mm are shown in Figs. 7(a)–7(c) and 7(a')–7(c'), respectively. It is clear in Figs. 6 and 7 that the experimental results included the diffraction pattern due to the singularity in the phase gradient in addition to the interference fringes, which confirms the calculated DIDs.

In this study, Fresnel diffraction due to a singularity in the phase gradient was investigated. It was shown that for a continuous phase distribution and discontinuous phase gradient, the Fresnel diffraction becomes significant. To illustrate this phenomenon, the DIDs for the FDM and FB were calculated using the Fresnel–Kirchhoff integral. In the

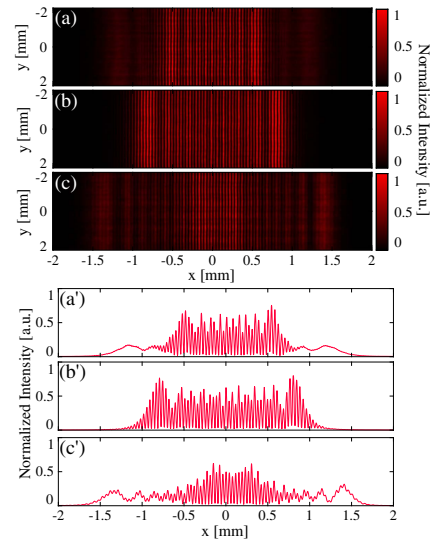


Fig. 7. DID and corresponding profiles for the FB: (a), (b), and (c) the DIDs and (a'), (b'), and (c') their corresponding profiles of the FB for $z = 70, 100$, and 160 mm .

geometrical overlapping area between two reflected beams from the FDM (two refracted beams from the FB), the intensity distribution was different from a common interference pattern, solely due to the diffraction effect. Furthermore, the intensity profile along the average propagation direction varied almost periodically. It was shown that by studying both the intensity variation along the average propagation direction and interference fringe spacing, the wavelength and the apex angle of the FDM (FB) were determined simultaneously. In addition, the theoretical results verified experimentally and the recorded DID were used to determine the wavelength and apex angle of FDM. The accuracy of the measurements was of the order of interferometry, and in addition, the wavelength of the incident light and apex angle were measured simultaneously.

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